Design, Simulation, Fabrication and Measurement of a 2×2 Corporate-Fed Microstrip Patch Antenna Array for S-Band LEO Satellite Reception

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1 Introduction

The exponential growth of wireless communication systems, radar technologies, and satellite applications has created an urgent demand for antennas that deliver high gain, directivity, and efficiency while remaining compact, lightweight, and cost-effective.

Traditional bulky antennas (e.g., parabolic dishes or horn arrays) often fail to meet the requirements of modern platforms such as unmanned aerial vehicles (UAVs), maritime vessels, portable IoT devices, and small satellites, where low profile, conformability, and ease of integration with RF circuitry are essential [1].

Microstrip patch antennas, popularised in the 1970s [2], address many of these constraints through their planar geometry, low fabrication cost, and compatibility with active circuits. A single rectangular microstrip patch offers a broadside radiation pattern but suffers from inherently low gain and narrow impedance bandwidth.

To overcome these limitations while preserving the core advantages of microstrip technology, planar antenna arrays with tapered corporate feeds are employed. The antenna array presented in this thesis is a 2.2 GHz tapered rectangular microstrip patch array utilising a 4-element configuration arranged in a square grid and fed via a corporate network incorporating linear tapered impedance transformers.

The design targets reception of S-band LEO satellite downlinks (telemetry, TT&C, or data from CubeSats/smallsats) operating in the 2200 MHz to 2290 MHz range, a band allocated by the ITU for deep-space and near-Earth research [3].

1.1 Project Motivation

The purpose of this project was to design, simulate, fabricate, and experimentally validate a microstrip patch antenna array that operates at approximately 2.2 GHz with sufficient bandwidth to cover the 2200 MHz to 2290 MHz spacecraft-to-Earth downlink band.

Table 1: Frequency Bands Allocated by the International Telecommunication Union (ITU)

Band Designation	Downlink for spacecraft within 2 million km (space to Earth)
S-band (MHz)	2200–2290

This antenna forms the core of a larger ground-station system known as the Cyberdeck, comprising a 4×4 antenna array interfaced with four analogue beamformers. These phase shifters on the analogue beamformer board steer at a degree interval of 22.5 degrees, requiring a half-power beam width of at least 22.5 degrees to ensure smooth beam scanning. The beamformed signals are processed digitally by a PlutoSDR, which decodes the satellite downlink and enables precise satellite tracking. To ensure frequency stability, the PlutoSDR is locked to a Leo Bodnar GPS-disciplined oscillator. System outputs are visualised on a display driven by a Raspberry Pi forming the Cyberdeck system.

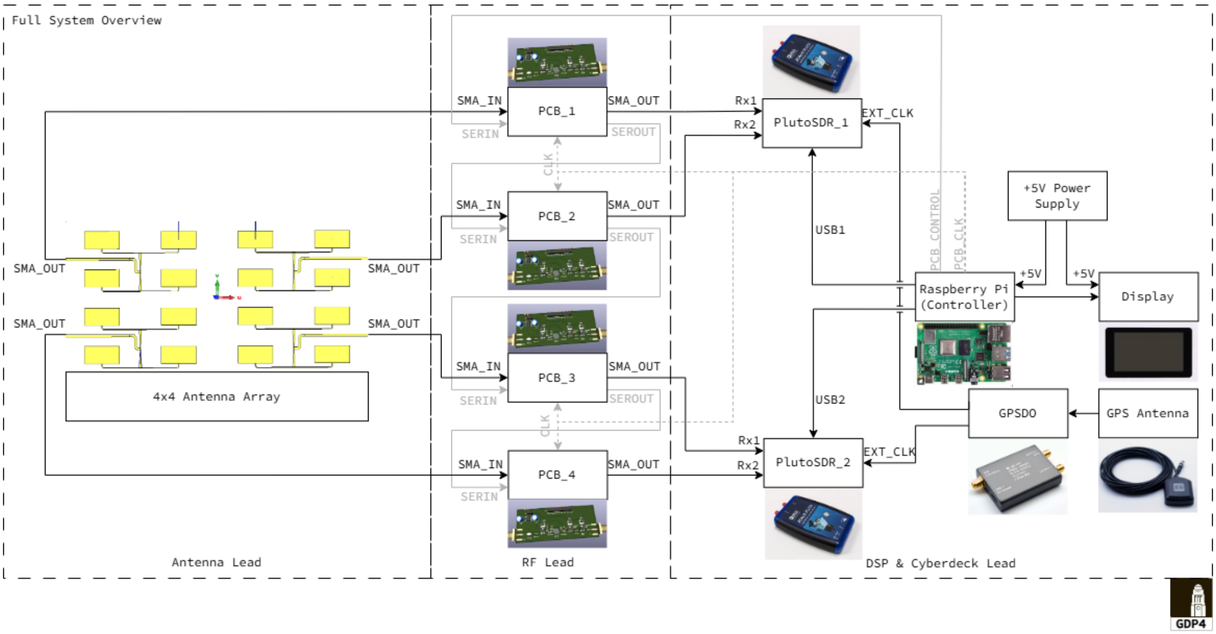


Figure 1: GDP Cyberdeck Overview

2 Antenna Theory and Principles

2.1 Antenna

An antenna is defined as a "that part of a transmitting or receiving system that is designed to radiate or to receive electromagnetic waves." [4] Antennas are transitional structures between free-space and a transmission line. The transmission line is used to transport electromagnetic energy from the transmitting source to the antenna or from the antenna to the receiver. [5]

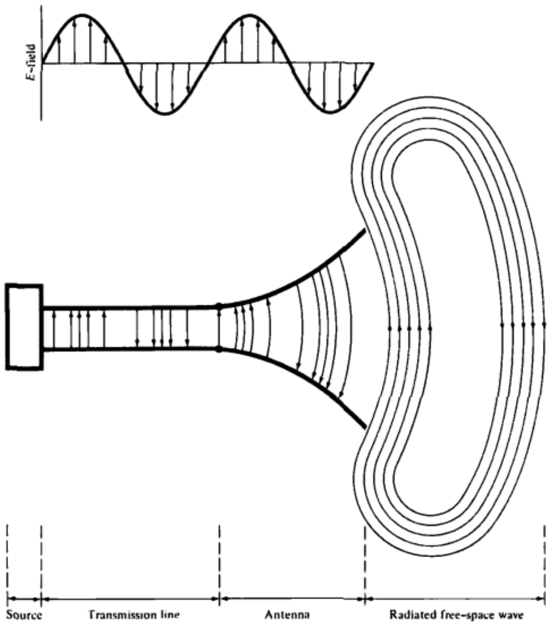


Figure 2: Antenna as a transition device

2.2 Radiation pattern (E-plane, H-plane, principal planes)

An antenna radiation pattern is defined as a mathematical function or a graphical representation of how an antenna radiates or receives electromagnetic energy as a function of direction in space. It shows the directional dependence of the radiated field strength, power density, or gain. For a linear polarized antenna, performance is often described in terms of its principal E- and H-plane patterns. [5]

2.3 Radiation Pattern Lobes

Radiation lobes are a "portion of the radiation pattern bounded by regions of weak radiation intensity also known as nulls." Various parts of a radiation pattern are referred to as lobes, which may be subclassified into major, minor, side, and back lobes. In most systems especially radar, low side lobe ratios are very important to minimize false target indications through the side lobes. [5]

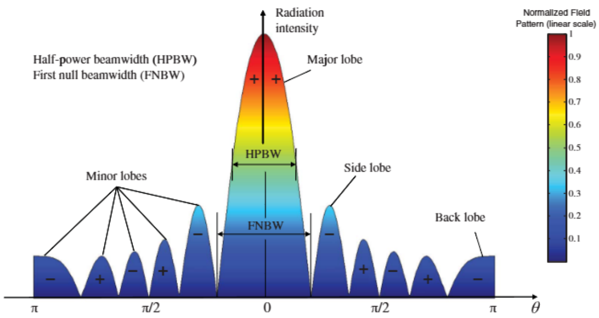


Figure 3: Radiation Lobes

2.4 Half-Power Beamwidth (HPBW)

The half-power beamwidth is defined as "In a plane containing the direction of the maximum of a beam, the angle between the two directions in which the radiation intensity is one-half the maximum value of the beam." The beam width of the antenna is a very important figure-of-merit, and is a trade-off with sidelobe level. Namely, as the beamwidth decreases, the sidelobe level increases, and vice versa. In addition, the beam width of the antenna is also used to describe the resolution capabilities of the antenna to distinguish between two sources is equal to half the first null beamwidth (FNBW/2), commonly approximated as the half-power beamwidth (HPBW). For a lot of antennas, the limiting factor is its impedance, and its bandwidth can be formulated in terms of Q, the quality factor.

2.5 Field Regions

The space surrounding an antenna is usually subdivided into three regions: (a) reactive near-field (b) radiating near-field (Fresnel) (c) far-field (Fraunhofer)

These regions are designated to identify the field structure in each. The reactive near-field region is defined as "that portion of the near-field region immediately surrounding the antenna wherein the reactive field predominates."

The radiating near-field (Fresnel) region is defined as "that region of the field of an antenna between the reactive near-field region and the far-field region wherein radiation fields predominate and wherein the angular field distribution is dependent upon the distance from the antenna. The inner boundary is taken to be the distance $R \geq 0.62\sqrt{D^2/\lambda}$ and the outer boundary is taken to be the distance $R < 2D^2/\lambda$ where D is the largest dimension of the antenna.

The far-field (Fraunhofer) region is defined as "that region of the field of an antenna where the angular field distribution is essentially independent of the distance from the antenna." If the antenna taken has a maximum overall dimension D, the far-field region is commonly taken to exist at distances greater than $2D^2/\lambda$ from the antenna. λ being the wavelength. [5]

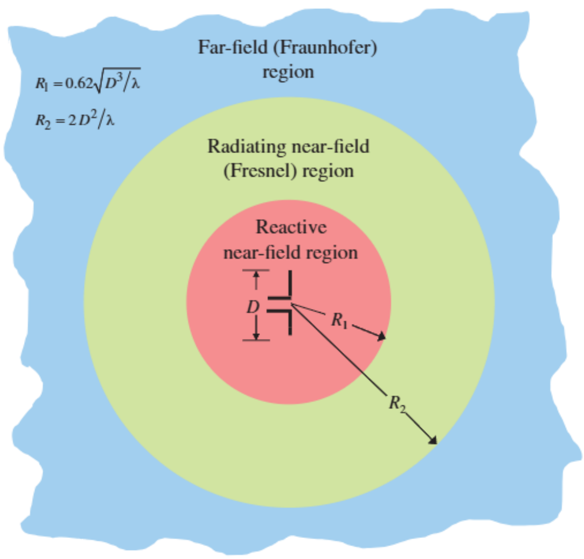


Figure 4: Field Regions

2.6 Directivity and gain (relationship via efficiency)

A useful measure describing the performance of an antenna is the gain and directivity. Directivity is a measure of how much the antenna concentrates radiated power in one direction relative to an isotropic radiator. It depends only on the shape of the radiaton pattern and assumes that all the power accepted by the antenna is actually radiated. Gain, on the other hand, accounts for the fact that some power accepted by the antenna is lost as heat (ohmic losses, dielectric losses). Realized gain, in addition to accounting for the transmission loss, accounts for the reflections due to impedance mismatch and polarization mismatches. [5]

2.7 The Quality Factor and Bandwidth

The bandwidth is an important parameter for antennas. It is closely linked to the *quality factor, Q*, which is a measure of how much lossless reactive energy is stored in a circuit compared to the average power dissipated. The bandwidth of an antenna is defined as "the range of frequencies within which the performance of the antenna, with respect to some characteristic, conforms to a specified standard." The bandwidth can be considered to be the range of frequencies on either side of a center frequency where the antenna characteristics (such as input impedance, pattern, beamwidth, gain, radiation efficiency) are within an acceptable value of those at the center frequency. A commonly used standard for bandwidth is the -10 dB impedance bandwidth.

2.8 Input impedance, return loss (S11)

Input Impedance is defined as "the impedance presented by an antenna at its terminals." It is expressed at the terminals as the ratio of the voltage to current or the ratio of the appropriate components of the electric to magnetic fields. When the antenna impedance Z_A is referred to the input terminals of the antenna, it reduces to input impedance.

2.9 Antenna Scattering Matrix Parameters

For measuring voltage and currents at microwave frequencies, direct measurements usually involve the magnitude (inferred from power) and phase of a wave traveling in a given direction or of a standing wave. In high frequency networks, equivalent voltages, currents, related impedance and admittance matrices, become an abstraction, with the ideas of incident, reflected, and transmitted waves, given by the scattering matrix. Like the impedance or admittance matrix for an N-port network, the scattering matrix provides a complete description of the network as seen at its *N* ports. While the impedance and admittance matrices relate the total voltages and currents at the ports, the scattering matrix relates the voltage waves incident on the ports to those reflected from the ports. The scattering parameters can be measured directly with a vector network analyzer. Once the scattering parameters of the network are known, conversion to other matrix parameters can be performed. [6]

The scattering matrix, or $[S]$ matrix, is defined in relation to these incident and reflected voltage waves as $[V^-] = [S][V^+]$, where V_n^+ is the amplitude of the voltage wave incident on port n and V_n^- is the amplitude of the voltage wave reflected from port n. The incident and emergent wave amplitudes on two terminal surfaces are related by the equations.

S-parameters are complex scalar quantities which specify the response of a two-port transducer at a given frequency. Once these quantities have been determined from measurements, it can be determined how this device will interact with others in a composite system.

The antenna is viewed as a two-port transducer which transforms wave amplitudes in a closed transmission line system to an angular spectrum of planes waves in the space system.

3 Design Methodology

3.1 Methods of Analysis

The design combines analytical models with full-wave electromagnetic simulation. The primary analytical approaches used are the transmission-line model (for initial sizing). Final dimensions and performance are validated and optimized using the full-wave solver in CST Microwave Studio.

The transmission-line model is the simplest and provides excellent physical insight, although it has limited accuracy for coupling effects. Full-wave methods are the most accurate and versatile, capable of handling mutual coupling, finite arrays, and complex feed networks, albeit with reduced physical insight.

3.1.1 Transmission-Line Model

The transmission-line model represents the rectangular microstrip patch as two radiating slots separated by a low-impedance transmission line of length L. It is used here as the starting point for determining patch dimensions, which are subsequently optimised in CST Studio Suite.

Fringing Effects and Effective Dielectric Constant Owing to the finite dimensions of the patch, the electric fields extend (fringe) beyond the physical edges into the air region. This fringing makes the patch electrically longer and wider. To account for the inhomogeneous dielectric (substrate + air), an effective relative permittivity ε_{eff} is introduced by [5]:

$$\varepsilon_{\text{eff}} = \frac{\varepsilon_r + 1}{2} + \frac{\varepsilon_r - 1}{2}\left(1 + 12\frac{h}{W}\right)^{-1/2}, \quad \frac{W}{h} > 1 \tag{1}$$

ε_{eff} is frequency-dependent but can be treated as constant near the design frequency for initial calculations.

The length extension due to fringing on each side is approximated by [5]:

$$\frac{\Delta L}{h} = 0.412\frac{(\varepsilon_{\text{eff}} + 0.3)\left(\frac{W}{h} + 0.264\right)}{(\varepsilon_{\text{eff}} - 0.258)\left(\frac{W}{h} + 0.8\right)} \tag{2}$$

The effective length of the patch then becomes

$$L_{\text{eff}} = L + 2\Delta L \tag{3}$$

Resonant Frequency For the dominant TM_{010} mode, the resonant frequency is given by [5]:

$$f_r = \frac{c}{2L_{\text{eff}}\sqrt{\varepsilon_{\text{eff}}}} \tag{4}$$

where $c = 3 \times 10^8$ m/s is the speed of light in free space.

Edge Input Impedance To eliminate the need for an inset or quarter-wave transformer, the patch is designed to present a $100\,\Omega$ input impedance at the radiating edge. The radiation resistance at the edge of a resonant rectangular patch is approximated as [6]:

$$Z_a \approx 90 \frac{\varepsilon_r^2}{\varepsilon_r - 1} \left(\frac{L}{W} \right)^2 \quad (\Omega) \tag{5}$$

Rearranging for the required aspect ratio gives

$$\frac{L}{W} = \sqrt{\frac{Z_a}{90} \frac{\varepsilon_r - 1}{\varepsilon_r^2}} \tag{6}$$

3.2 Step-by-Step Design Process

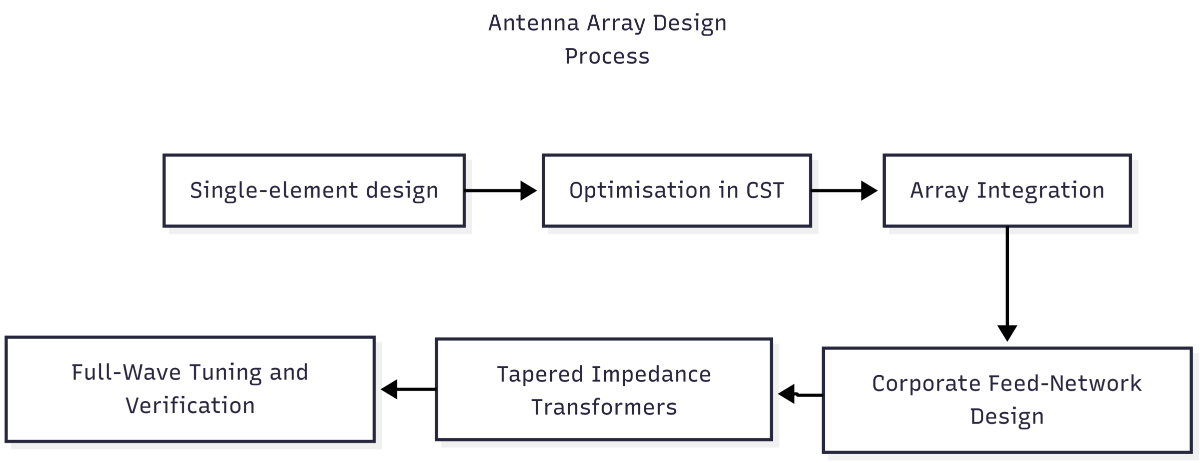


Figure 5: Antenna Array Design

Design parameters:

- Target resonant frequency: $f_r = 2.2\,\text{GHz}$
- Substrate: FR4 ($\varepsilon_r = 4.3$, $h = 1.6\,\text{mm}$, $\tan\delta = 0.025$)
- Target edge impedance: $Z_a = 100\,\Omega$
- Free-space wavelength: $\lambda_0 = 136.36\,\text{mm}$

3.3 Single-Element Patch Design

Using Eq. (6) with $Z_a = 100\,\Omega$:

$$\frac{L}{W} = \sqrt{\frac{100}{90} \frac{(4.3-1)}{4.3^2}} = 0.4453$$

Initial length (ignoring fringing), using Eq. (4) with ε_r instead of ε_{eff}:

$$L = \frac{3 \times 10^8}{2 \times 2.2 \times 10^9 \sqrt{4.3}} = 32.88\,\text{mm}$$

Corresponding width:

$$W = \frac{32.88}{0.4453} \approx 73.84\,\text{mm}$$

Now including fringing effects. First compute the effective permittivity using Eq. (1):

$$\varepsilon_{\text{eff}} = \frac{4.3+1}{2} + \frac{4.3-1}{2}\left(1+12\frac{1.6}{73.84}\right)^{-1/2} \approx 4.12$$

Effective length using Eq. (4):

$$L_{\text{eff}} = \frac{3\times10^8}{2\times2.2\times10^9\sqrt{4.12}} \approx 33.59\,\text{mm}$$

Fringing extension ΔL using Eq. (2):

$$\Delta L = 0.412\times1.6\times\frac{(4.12+0.3)\left(\frac{73.84}{1.6}+0.264\right)}{(4.12-0.258)\left(\frac{73.84}{1.6}+0.8\right)} \approx 0.71\,\text{mm}$$

Final physical patch length using Eq. (3):

$$L = L_{\text{eff}} - 2\Delta L \approx 33.59 - 2\times0.71 = 32.17\,\text{mm}$$

These dimensions were used as the initial guess and subsequently optimised in CST (final optimised size after accounting for feed effects and mutual coupling: 31×62 mm).

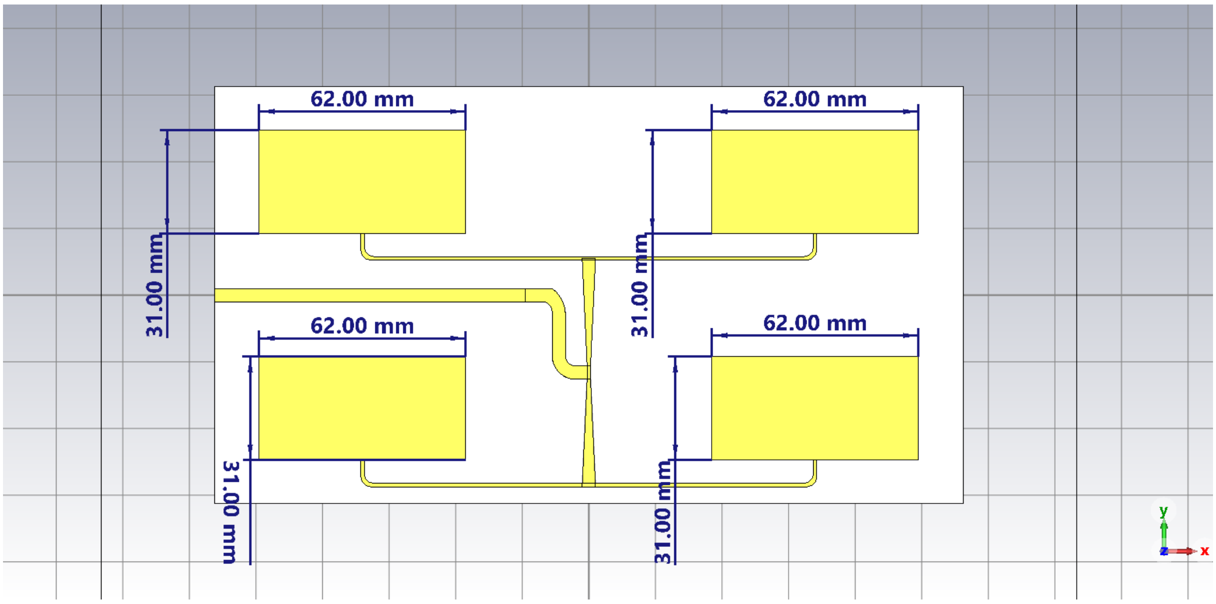


Figure 6: Antenna Dimensions

3.4 Antenna Array Integration (2×2)

3.4.1 Inter-Element Spacing

An array with its peak at θ_0 can have other peak values subject to the choice of spacing d_x. Such peaks are called grating lobes and are shown to occur at angles θ_p such that

$$sin\theta_p = sin(\theta_0) + \frac{p\lambda}{d_x}, \qquad p = \pm1, \pm2, \ldots \tag{7}$$

To suppress grating lobes while maintaining high directivity and manageable mutual coupling, a criterion for determining the maximum element spacing for an array scanned to a given scan angle θ_0 at frequency f is to set the spacing so that the nearest grating lobe is at the horizon.

This satisfies the grating-lobe condition given in [7]

$$d \leq \frac{\lambda_0}{1+|\sin\theta_{\max}|} \tag{8}$$

Thus the centre-to-centre spacing is chosen as $d_x = d_y = 0.5\lambda_0 = 68\,\text{mm}$, working for scan angles up to $\pm 90^\circ$ (theoretical limit) and is suitable for broadside operation.

3.4.2 Element Positions

For the 2×2 array (origin at array centre):

$$(x_1, y_1) = (-68, -34)\,\text{mm}, \qquad (x_2, y_2) = (+68, -34)\,\text{mm},$$
$$(x_3, y_3) = (-68, +34)\,\text{mm}, \qquad (x_4, y_4) = (+68, +34)\,\text{mm}.$$

3.5 Corporate Feed Network Design for Array Integration

A corporate (parallel) feed network based on microstrip T-junction power dividers is employed to deliver equal-amplitude, in-phase excitation to every element of the array. Each T-junction provides a 3 dB (1:2) power split.

The single rectangular patch is edge-fed and designed to present an input impedance of exactly $100\,\Omega$ at the feed point. The characteristic impedance Z_0 of the microstrip lines on the FR4 substrate ($\varepsilon_r = 4.3$, $h = 1.6\,\text{mm}$) is determined from the width-to-height ratio W/h using the Hammerstad–Jensen approximations [8]:

For narrow lines ($\frac{W}{h} \leq 1$):

$$Z_0 = \frac{60}{\sqrt{\varepsilon_{\text{eff}}}} \ln\left(\frac{8h}{W} + \frac{W}{4h}\right) \tag{9}$$

For wide lines ($\frac{W}{h} \geq 1$):

$$Z_0 = \frac{120\pi}{\sqrt{\varepsilon_{\text{eff}}}\left[\frac{W}{h} + 1.393 + 0.667\ln\left(\frac{W}{h} + 1.444\right)\right]} \tag{10}$$

These quasi-static expressions require the effective relative permittivity ε_{eff}, which is itself a function of W/h and ε_r and is obtained iteratively. Numerical solution of Eqs. (9)–(10) at the design frequency $f = 2.2\,\text{GHz}$ yields the strip widths and corresponding effective permittivities shown in Table 2.

Target Impedance	**Strip Width** W	ε_{eff}
$50\,\Omega$	3.15 mm	3.27
$100\,\Omega$	0.71 mm	2.98

Table 2: Microstrip line dimensions on FR4 substrate ($\varepsilon_r = 4.3$, $h = 1.6\,\text{mm}$).

3.5.1 Tapered Impedance Transformers

To achieve broadband matching between feed-network stages, linear tapered transmission-line transformers are used instead of discrete quarter-wave sections. A continuous taper provides a smooth impedance transition that minimises reflections over a wide frequency range. The impedance profile of a linear taper is given by [9]:

$$Z_0(z) = Z_S + (Z_L - Z_S)\frac{z}{L}, \qquad 0 \le z \le L \tag{11}$$

where Z_S and Z_L are the source and load impedances, respectively, and L is the physical length of the taper. Performance improves monotonically with increasing electrical length because reflections have more distance over which to cancel.

3.5.2 Impedance Transformation Steps in the Feed Network

The corporate feed network (illustrated in Fig. 7) is implemented as follows:

1. Each patch is edge-matched to $100\,\Omega$ and fed by a $100\,\Omega$ microstrip line ($W \approx 0.71\,\text{mm}$).
2. **First combining stage (two patches):** Two $100\,\Omega$ lines combine in parallel at a T-junction, yielding
$$Z_{\parallel 1} = \frac{100\,\Omega}{2} = 50\,\Omega. \tag{12}$$
3. **Second combining stage (two pairs of patches):** Each $50\,\Omega$ branch is transformed to $100\,\Omega$ using a linear tapered section (Eq. (11)). The two resulting $100\,\Omega$ lines then combine in parallel at the next T-junction, producing the final input impedance
$$Z_{\text{in}} = \frac{100\,\Omega}{2} = 50\,\Omega. \tag{13}$$
4. A final $50\,\Omega$ microstrip line ($W \approx 3.15\,\text{mm}$) connects the network to the edge-mounted SMA connector.

This architecture ensures a $50\,\Omega$ system impedance at the input while maintaining equal power division and phase coherence across all elements. The tapered transformers (Eq. (11)) provide significantly wider impedance bandwidth than conventional stepped or quarter-wave matching networks.

3.6 Simulated Performance Characteristics

This section presents the simulated performance of the designed 2×2 and 4×4 microstrip antenna arrays. The radiating elements are rectangular patches with optimised dimensions (length = 31 mm, width = 62 mm) on an FR4 substrate ($\varepsilon_r = 4.3$, thickness $h = 1.6\,\text{mm}$, loss tangent $\tan\delta = 0.025$). The patch and ground-plane metallisation is 35 μm copper. These final patch dimensions differ from the initial theoretical values because of full-wave optimisation performed in CST Studio Suite to account for mutual coupling, feed-network effects, and substrate dispersion.

The feeding network is a corporate (parallel) microstrip structure that utilises T-junction power dividers and linear tapered impedance transformers. The special feature of the design is the use of tapered transformers, which provide a wider impedance bandwidth than conventional quarter-wave or inset-feed matching techniques. All simulations were performed in CST Studio Suite. Key performance metrics evaluated include return loss (S_{11}), impedance bandwidth, realised gain, directivity, radiation efficiency, and voltage standing-wave ratio (VSWR).

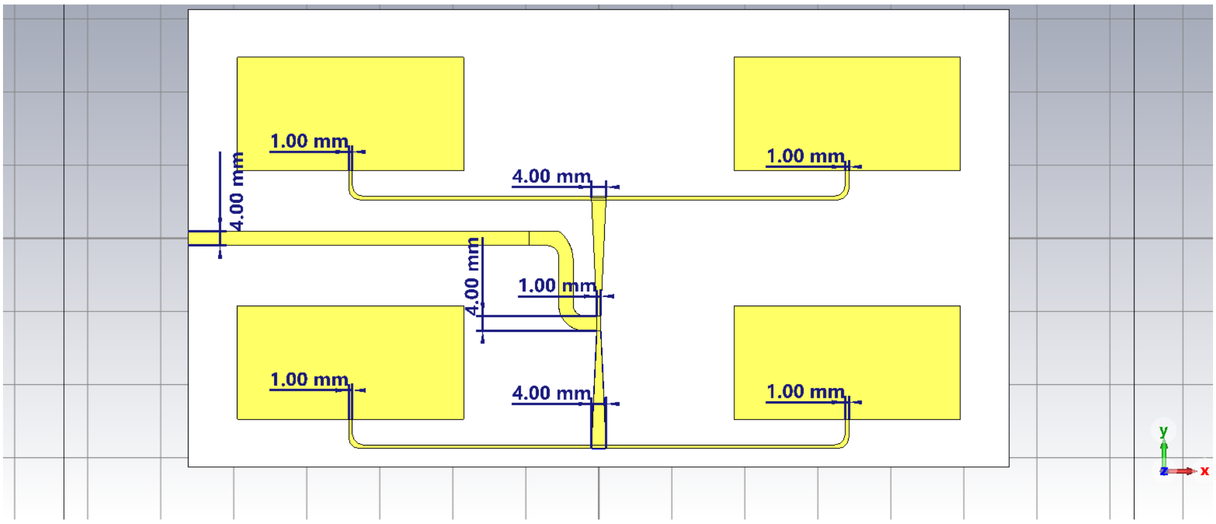


Figure 7: Corporate feed network with linear tapered impedance transformers (all dimensions in mm).

4 Simulated Performance Characteristics

4.1 Introduction

This section presents the simulated performance of the designed **2×2** and **4×4** antenna arrays.

The patches are rectangular, with individual patch dimensions (length = 31 mm, width = 62 mm), substrate material (FR-4 with a dielectric constant of 4.3, thickness of 1.6 mm, loss tangent of 0.025), the patch and ground plane material is copper annealed with thickness of 0.035 mm.

The patch dimensions are different from the calculated dimensions following optimizations in CST Studio.

The feeding method is a microstrip line and its feed network is a corporate feed and a power divider with tapered impedance transformers.

The special feature of the antenna is its tapered impedance transformer which enables a wider bandwidth impedance match than conventional quarter-wave or inset impedance transformers.

Simulations were performed in CST. Key performance metrics include return loss (S_{11}), impedance bandwidth, gain, directivity, radiation efficiency, and VSWR.

4.2 Antenna Array Geometries

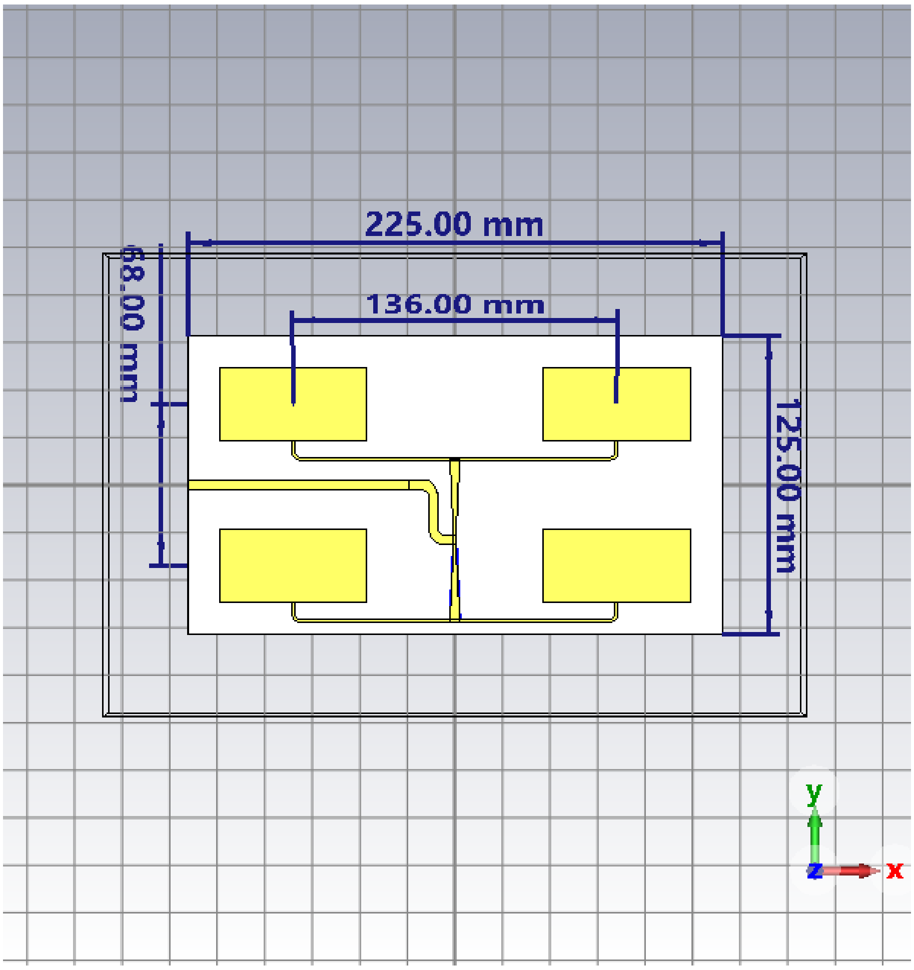
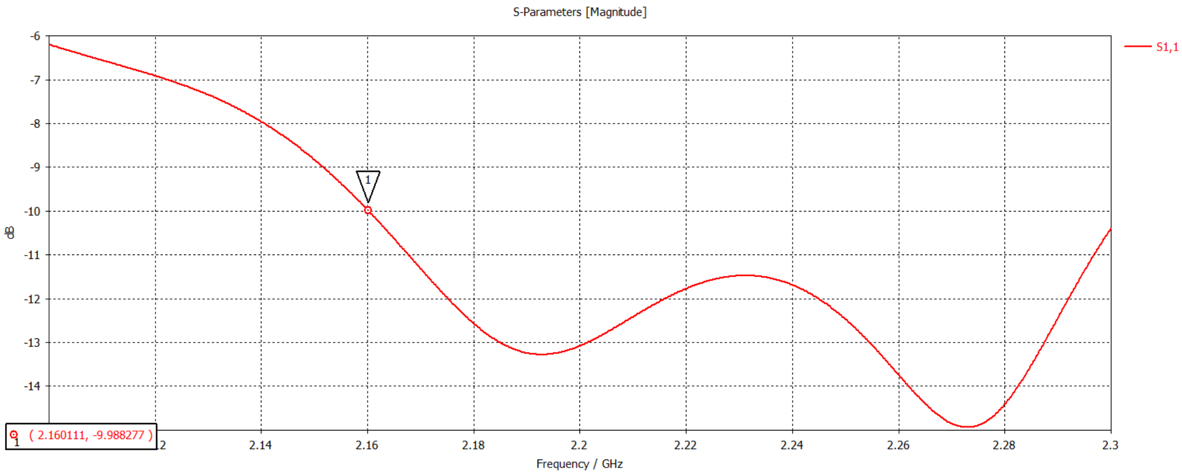


Figure 8: Geometry of the 2×2 antenna array (element spacing = 0.5 λ)

4.3 Detailed Results for 2×2 Antenna Array

The 2×2 array achieves good impedance matching with S_{11} below -10 dB across the band of interest.



(a) Simulated S_{11} (return loss)

Figure 9: Key frequency-domain characteristics of the 2×2 array

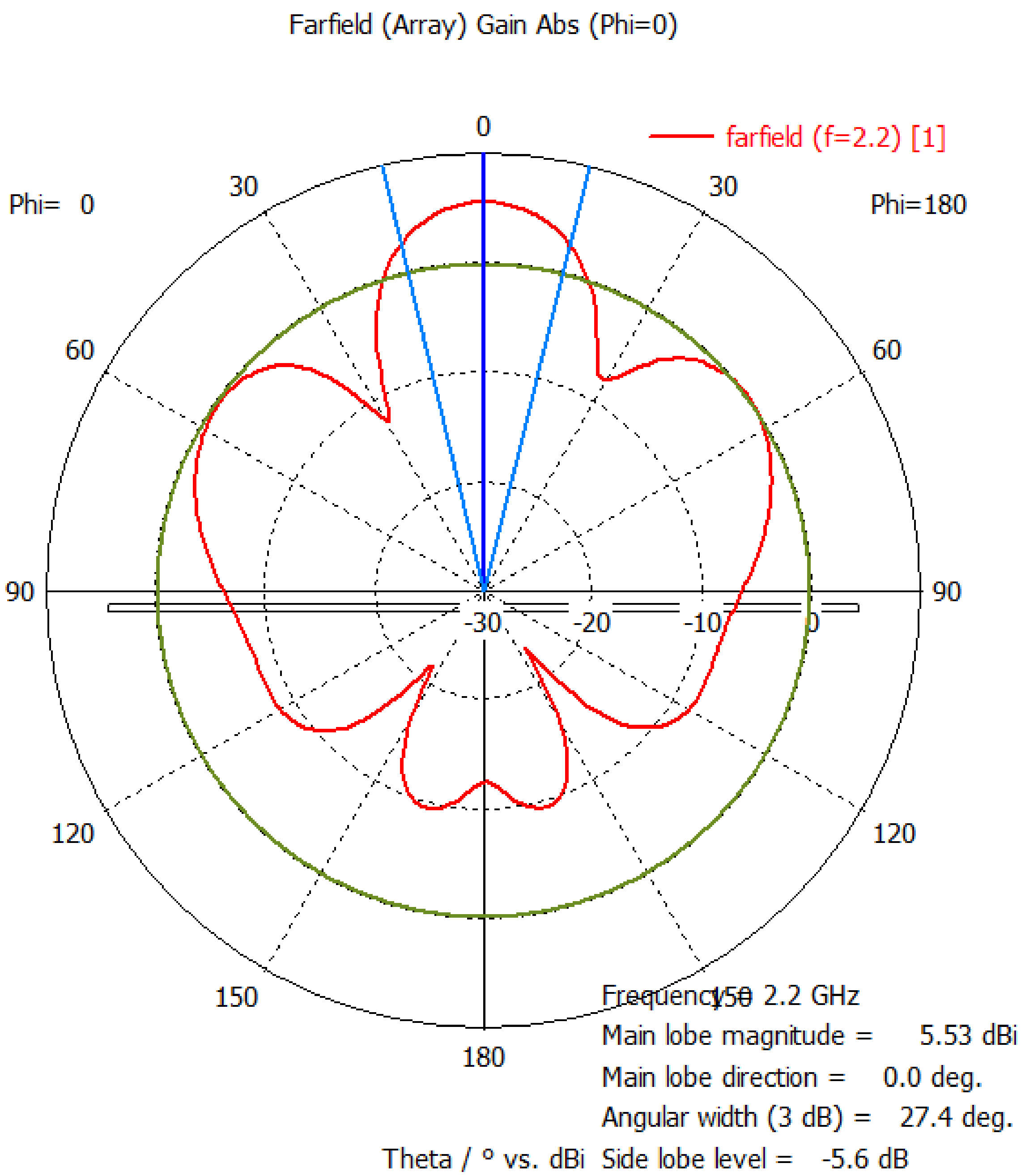


Figure 10: Simulated farfield gain radiation pattern plots of the 2x2 array at 2.2 GHz

4.4 Performance Comparison

The following table summarises the main simulated performance characteristics at the operating frequency: 2.2 GHz.

Table 3: Simulated Performance Characteristics

Parameter	**2×2 Array**
Resonant Frequency	2.2 GHz
Return Loss (S_{11})	-17 dB
-10 dB Impedance Bandwidth	140 MHz (6.1 %)
Peak Gain	5.53 dBi
Half-Power Beamwidth (E-plane)	27.4°
Side-lobe level	-5.6 dBi

5 Fabrication and Implementation

5.1 Substrate Material Selection

FR4 (flame-retardant grade 4) is an epoxy-glass composite. FR4 is a heterogeneous material with manufacturers producing many variants.

There are some key reasons for the differences in the dielectric constant (Dk).

One of the reasons is the glass-to-resin volume V_{rsn}: Electrical glass has a higher Dk (5.6-6.5) than the epoxy resin (3.2-4.0). More resin lowers the effective Dk. Different glass-weave styles (e.g. 106, 2116, 7628) change the resin content and create local anisotropy ("fibre-weave effect"), leading to Dk variations across a single board. [10]

Another reason is the frequency dependence: Dk decreases at higher frequencies due to reduced polarization response. [11]

5.2 Surface Finish Selection

An Electroless Nickel Immersion Gold (ENIG) surface finish was chosen over Hot Air Solder Leveling (HASL, both lead-free and leaded variants). This decision was driven by RF performance requirements at 2.2 GHz.

At microwave frequencies, the skin effect confines current flow to the conductor surface. Surface roughness increases the effective current path length, elevating conductor losses and reducing radiation efficiency. ENIG produces a significantly smoother and more uniform surface than HASL, thereby minimising roughness-induced attenuation.

Additionally, ENIG offers excellent thickness uniformity (typically 3–6 μm nickel + 0.05–0.2 μm gold), which helps preserve precise characteristic impedance control across the corporate feed network and radiating patches. In contrast, HASL often results in uneven solder deposits that introduce local variations in effective dielectric constant and impedance, leading to unwanted reflections.

ENIG also provides superior long-term oxidation resistance via its thin gold layer, ensuring stable electrical performance and reliability of the RF traces and patch elements over time. [12]

5.3 SMA Connector Mounting

Subminiature version A (SMA) connectors are widely used in RF and microwave systems to interface coaxial cables with printed circuit boards and other components. They are available in various configurations, including straight, right-angle, bulkhead, edge-mount, and PCB-mount variants. [13]

In this work, the SMA connector is not explicitly modelled in the electromagnetic simulations. According to Hirano et al. [14], modelling only the inner coaxial structure is sufficient when no significant electromagnetic fields exist outside the connector. Full modelling of the exterior geometry and receptacle is only necessary when external fields are present. As the fields are confined within the coaxial transition in this design, the simplified coaxial approximation was adopted. This simplification reduces computational complexity while maintaining acceptable accuracy.

For physical implementation, the CON-SMA-EDGE-S-ND edge-mount SMA connector (RF Solutions) was selected.

Signal Routing As there 4 different ground pins, two on the bottom, and two on the top, and only one signal pin on the top, it is of high importance that the ground pins be isolated from the signal isolated otherwise the connector will be shorted to ground.

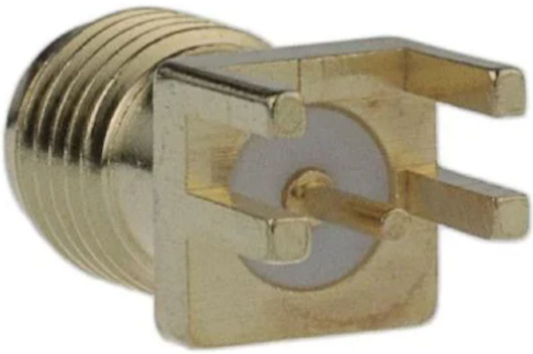


Figure 11: SMA Mount

Here is an example of an SMA edge mount being shorted to ground. There will be no signal transmitted or received through this connector.

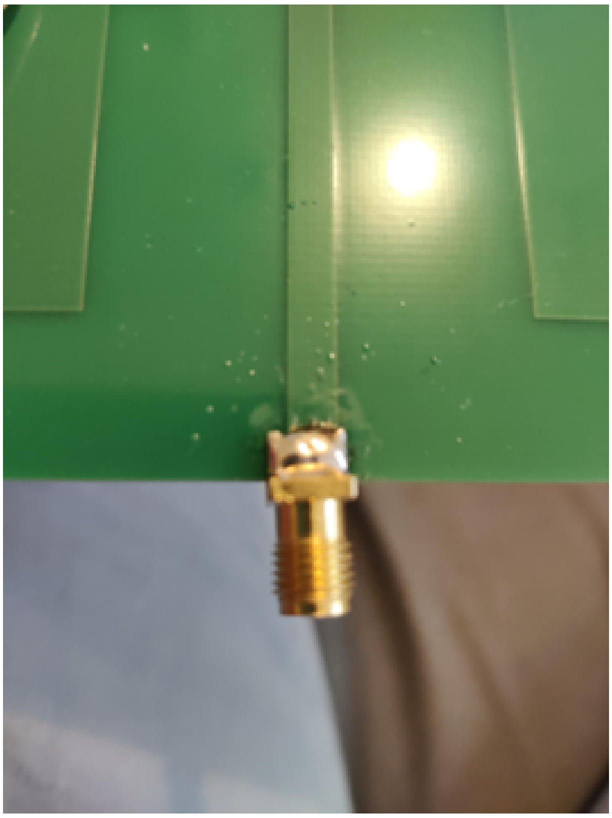


Figure 12: Shorted SMA Mount

Here is an example of the SMA edge mount correctly being able to transmit and receive signals. This is achieved by isolating the ground pins from the signal pin, in this case, the 2 top ground pins were trimmed off to avoid any risk of the signal pin being shorted.

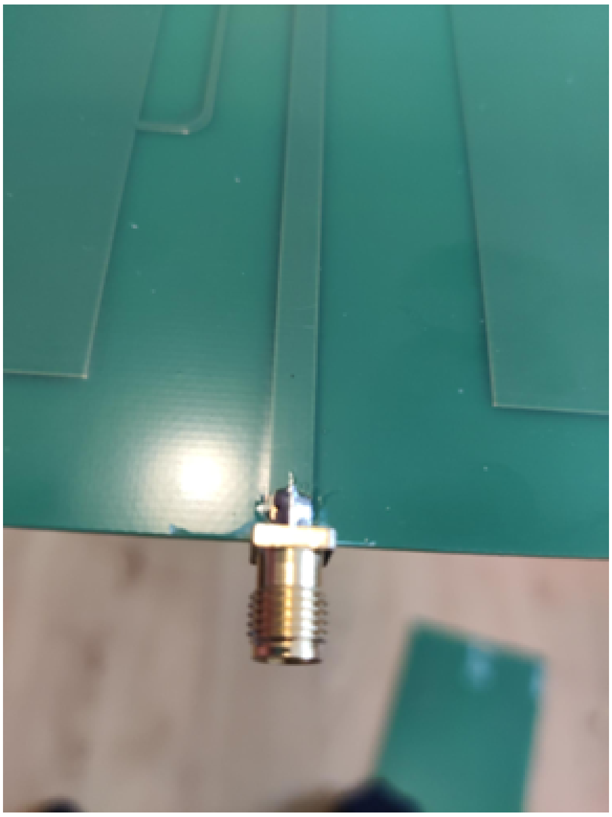


Figure 13: Corrected SMA Mount

5.4 Types of Calibration on the FieldFox N9923A

The FieldFox N9923A supports three main calibration categories: **CalReady** (factory), **QuickCal**, and **Mechanical Cal**.

5.4.1 Calibration Options

- **CalReady**: Factory 2-port calibration performed at the Type-N test ports. Traceable to NIST standards. Always active for direct DUT connections. Corrects all 12 systematic error terms. Typically more accurate than low-cost mechanical standards.
- **QuickCal** (Option 112): Built-in electronic calibration. Fast, kit-free, and ideal for field use. Corrects for temperature drift and added cables/adapters with minimal operator interaction.
- **Mechanical Cal**: Traditional calibration using discrete Open, Short, Load, and Thru standards. Requires a high-quality calibration kit. Offers maximum accuracy when premium standards with characterized coefficients are used.

Electronic calibration (QuickCal or ECal) reduces operator error compared to mechanical methods and provides NIST-traceable results with fewer connections. [15]

5.4.2 Mechanical Calibration Standards

Mechanical calibration uses precision Open/Short/Load (and Thru) standards. Accuracy depends on standard quality and connection repeatability.

Keysight kits provide electrical models including:

- Open: fringing capacitance
- Short: inductance
- Load: resistance and offset delay
- Thru: delay and loss

Cheap or generic standards are often treated as ideal (zero-length), which introduces larger residual errors. A proper characterized Type-N kit is recommended for best results. [15]

5.4.3 Comparison

Type	Kit Required	Best For
CalReady	No	Direct port connections
QuickCal	No	Field measurements, drift correction
Mechanical	Yes	Highest precision (with quality kit)

For the reported work, coarse mechanical (no Type-N kit), Open/Short/Load calibration was performed. Phase-stable cables and careful connections are essential for good repeatability.

6 Experimental Validation

6.1 1D Measurements, Input Reflection, Impedance Bandwidth

The fabricated prototypes were characterised using a Keysight FieldFox N9923A Vector Network Analyser (VNA).

Figure 14: FieldFox N9923A

Two 2 m low-loss RG316 cables were used for transmit and receive paths.

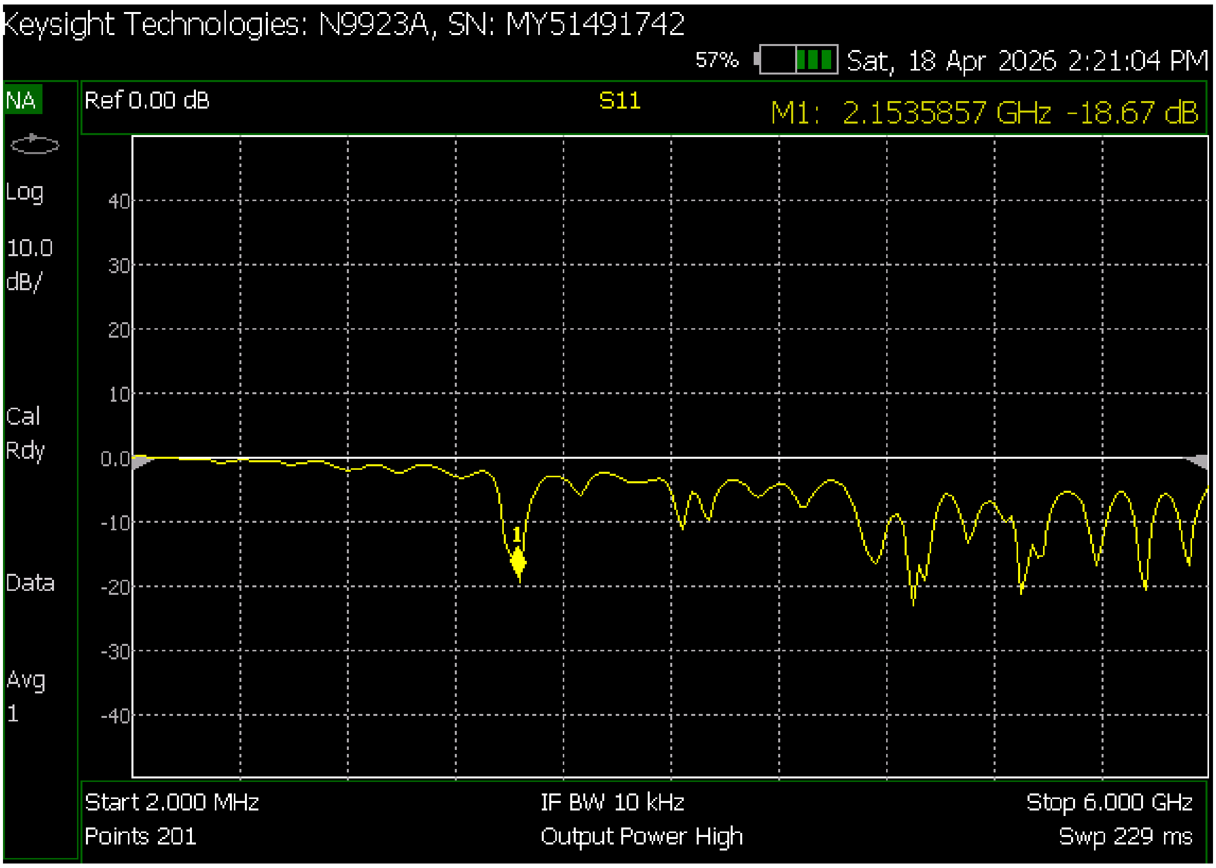


Figure 15: Measured and simulated reflection coefficient S_{11} of the antenna array.

The measured resonance occurs at 2.153 GHz with $S_{11} = -18.67$ dB, 117 MHz below the simulated resonance.

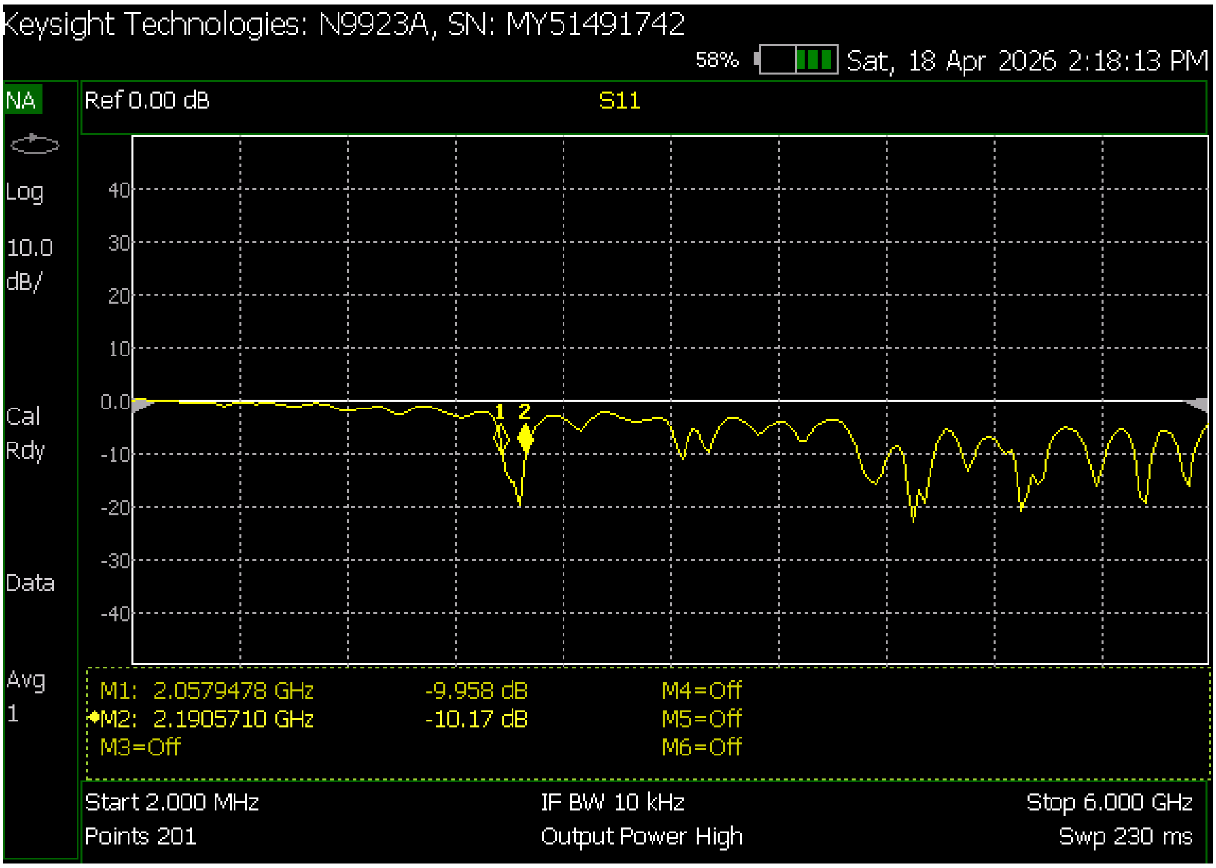


Figure 16: Measured -10 dB impedance bandwidth.

The -10 dB impedance bandwidth is 132 MHz, in close agreement with the simulated value of 140 MHz.

6.2 Dielectric Constant

The effective dielectric constant of a PCB material depends on several factors, including test method, trace geometry, operating frequency, resin content, and material thickness.

Particularly important is the resin-to-glass weave pattern. Most PCB laminates are inhomogeneous and anisotropic due to the glass fiber weave. This leads to fiber weave effects such as differential skew and cavity resonances. [16]

In this work, a 117 MHz downward shift in resonant frequency was observed between simulation and measurement. Since all other parameters were identical, the discrepancy was attributed to the use of an incorrect dielectric constant in the original simulation.

The resonant frequency of a rectangular microstrip patch antenna is approximately given by

$$f_r \approx \frac{c}{2L\sqrt{\varepsilon_r}},$$

which implies

$$f_r \propto \frac{1}{\sqrt{\varepsilon_r}}.$$

With fixed patch dimensions, the actual dielectric constant can be estimated as

$$\varepsilon_{r,\text{actual}} \approx \varepsilon_{r,\text{design}} \times \frac{f_{\text{design}}}{f_{\text{measured}}}.$$

Substituting the values:

$$\varepsilon_{r,\text{actual}} \approx 4.3 \times \frac{2.27\,\text{GHz}}{2.15\,\text{GHz}} \approx 4.8.$$

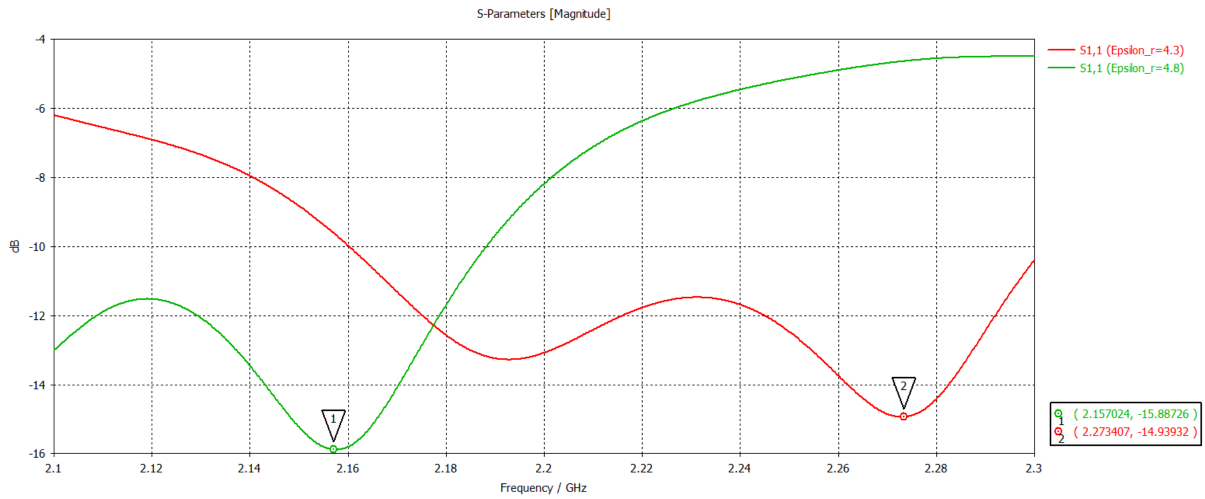


Figure 17: Extracted dielectric constant adjustment

The simulation was repeated using $\varepsilon_r = 4.8$, resulting in significantly improved agreement with the measured resonance.

6.3 Far-Field Radiation Pattern Measurements

6.3.1 Measurement Setup and Procedure

All far-field measurements were performed in an indoor range using the three-antenna absolute-gain method. The FieldFox N9923A VNA served as both transmitter and receiver.

Far-Field Distance The minimum far-field distance was verified using the criterion $R > 2D^2/\lambda$, where D is the largest antenna dimension (≈ 0.23 m for the 2×2 array). At 2.15 GHz ($\lambda \approx 0.139$ m), the far-field region begins at approximately 0.76 m. All measurements were performed at $R = 0.8$ m.

System Calibration A full two-port VNA response calibration was performed using a short, open, and load calibration references. Cable and connector losses were quantified via a direct thru measurement.

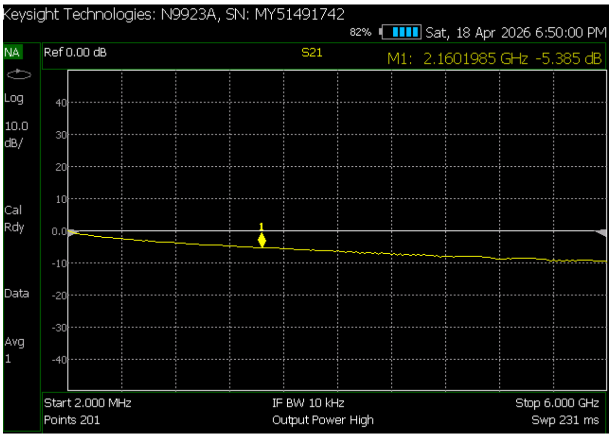


Figure 18: Measured cable and connector losses (thru calibration).

The measured cable and connector loss measured is -5.385 dB.

Boresight The antenna boresight is the central axis of the main beam - the direction in which the antenna is pointing when it has maximum or radiation intensity.

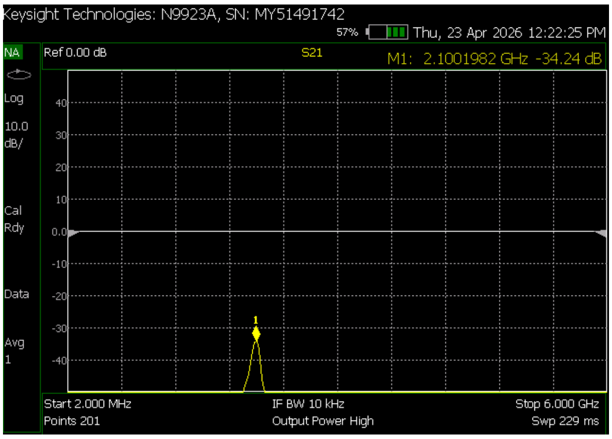


Figure 19: Antenna Boresight

There is a 53 MHz frequency shift between the best S11 (impedance match/resonance) and the peak S21 (boresight transmission/gain).

Maximum gain does not necessarily coincide with the best impedance match, because gain depends on necessarily coincide with the best impedance match, because absolute gain depends on radiation efficiency + directivity, while S11 is purely about power acceptance at the feed. [17]

Beam / polarization alignment There are two commonly prescribed methods for aligning antennas. One depends on the mechanical properties of the antenna while the other depends on its electrical properties. The goal of the alignment procedure is to align the mechanical or electrical boresights of the transmitting and receiving antennas with the reference z-axis of the extrapolation range and to provide a reference for the polarization measurements. [18]

Mechanical alignment Mechanical alignment works by aligning basing on the physical structure of the antennas. This is achieved using levels, transits, lasers, reference marks on the antenna body/flange, or optical tools to ensure the mechanical boresight points exactly along the range centerline. [18]

Electrical alignment Electrical alignment works by aligning based on the actual RF performance. A signal is transmitted between the antennas and azimuth/elevation is slightly tweaked until you get the maximum received power/signal. This peaks the electrical boresight - the real direction of peak gain/radiation. It ignores any small manufacturing offsets between the physical body and the true RF beam. [18]

Both mechanical and electrical alignment procedures were employed. Mechanical alignment was used to establish the range axis. Electrical alignment was then refined by maximising received power through small azimuth and elevation adjustments. The measurement sequence is repeated for all three pairs of antennas to provide the necessary data for the solution of the three antenna equations.

6.4 Gain Measurement Using the Three-Antenna Method

Accurate determination of antenna gain is essential for the evaluation of communication, radar, and other wireless systems. Prior to the mid-1970s, the most common approach was the gain substitution (or gain-comparison) method, in which the response of the test antenna was compared against a known standard antenna (typically a pyramidal horn). However, uncertainties arising from fabrication tolerances and theoretical approximations led the National Bureau of Standards (NBS) to favour "standard measurement methods" rather than "standard antennas." These methods allow accurate calibration of an arbitrary antenna without requiring a precisely characterised reference antenna. [17] [18]

The three-antenna method provides an absolute gain measurement without requiring a calibrated reference antenna. It is based on the Friis transmission equation expressed in terms of the measured transmission coefficient:

$$S_{21}(\mathrm{dB}) = G_t(\mathrm{dB}) + G_r(\mathrm{dB}) - \mathrm{FSPL}(\mathrm{dB}) - L_c(\mathrm{dB}) \tag{14}$$

where $\mathrm{FSPL} = 20\log_{10}\left(\frac{4\pi R}{\lambda}\right)$ is the free-space path loss and L_c is the total cable/connector loss.

Three antennas (labelled Ant1, Ant2, and Ant3) were measured in pairs at the same distance and frequency, with all antennas aligned to boresight.

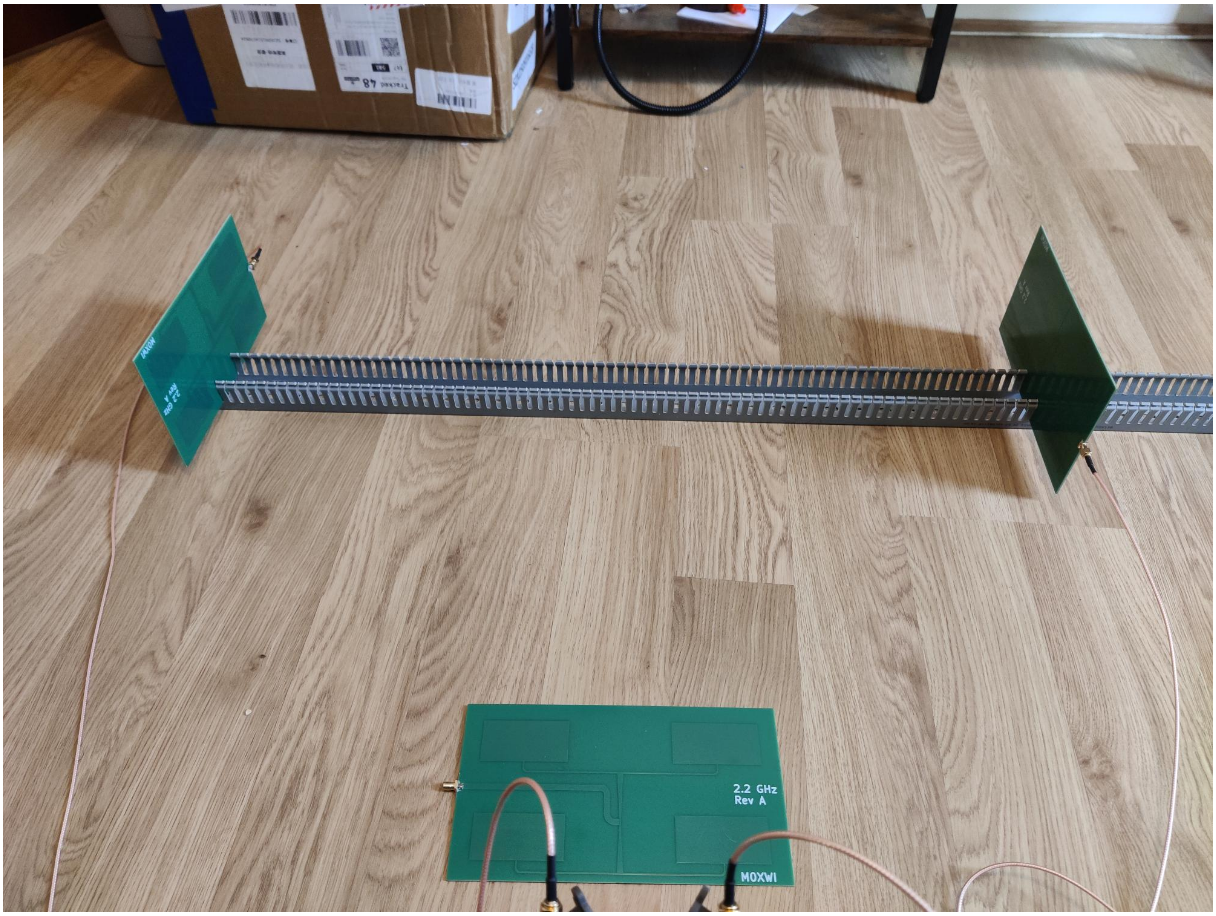


Figure 20: Three-antenna method measurement setup.

Measurement	A_1 (Ant1–Ant2)	A_2 (Ant1–Ant3)	A_3 (Ant2–Ant3)
S_{21} (dB)	-36.45	-35.87	-35.97

Table 4: Three-antenna method transmission measurements at 2.153 GHz.

The free-space path loss at $R = 0.8\,\text{m}$ and $\lambda \approx 0.139\,\text{m}$ is

$$\text{FSPL} = 20\log_{10}\left(\frac{4\pi \times 0.8}{0.139}\right) \approx 37.16\,\text{dB}.$$

Cable losses from the thru measurement were $L_c = 5.385\,\text{dB}$, giving a total path loss $PL = 42.55\,\text{dB}$. Solving the system of equations yields the gains of the three antennas:

$$G_1 = \frac{A_1 + A_3 - A_2 + PL}{2} \approx 5.90\,\text{dBi},$$
$$G_2 = \frac{A_1 + A_2 - A_3 + PL}{2} \approx 6.10\,\text{dBi},$$
$$G_3 = \frac{A_2 + A_3 - A_1 + PL}{2} \approx 7.06\,\text{dBi}.$$

The Antenna Under Test (AUT) gain is taken as the average of the relevant measurements (G_1, G_2, G_3).

$$G_{avg} = \frac{5.9 + 6.1 + 7.06}{3} = 6.353\,dBi$$

6.4.1 Half-Power Beamwidth (HPBW)

Figure 21: HPBW measurement setup.

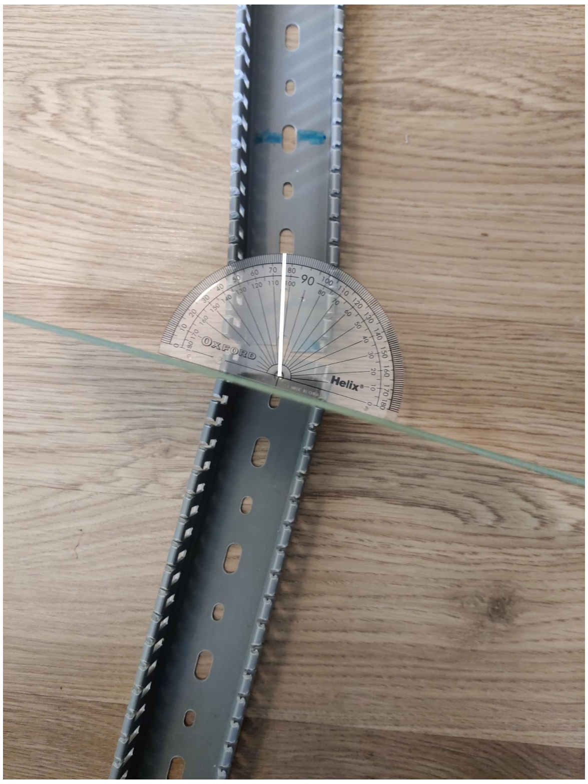


Figure 22: HPBW Angle

Half power beam width, HPBW, is the angular width (in degrees) of the main beam. They are determined from the -3 dB points on the normalised radiation patterns in both principal planes. It is measured in the E-plane (the plane containing the electric field vector, usually the narrow plane for rectangular plane for rectangular patches). The angular -3 dB width appears to be 15 degrees (90-75 degrees). This would be stepped beam scanning if using the 22.5 degrees phase shifters, which would lead to less effective satellite tracking. There is a significant difference of 12 degrees between the measured and simulated beamwidths. This would be likely due to the coarse measurement tools available, multipath and reflections.

6.4.2 Side-Lobe Level (SLL)

Figure 23: Side-lobe level measurement setup.

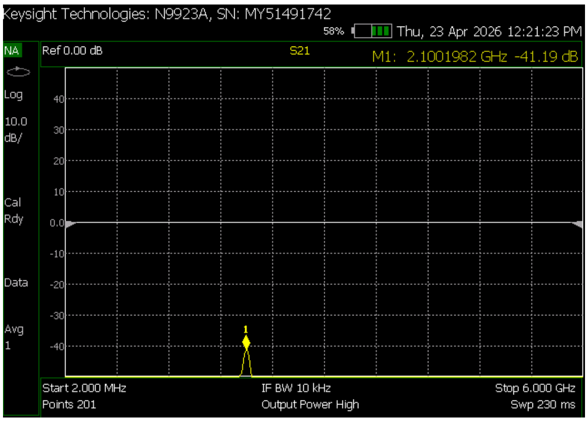


Figure 24: Measured side-lobe level.

The side-lobe level is defined as the difference (in dB) between the main-lobe peak and the highest side lobe.

- Formula: $SLL(dB) = P_{front} - P_{SLL}$

Lower SLL is better at it means less unwanted radiation in off-axis directions. $SLL = -34.24 - (-41.19) = 6.95\ dB$

6.5 Assumptions and Sources of Error

The standard far-field three-antenna method relies on the following key assumptions:

1. Frequency stability of the entire measurement system,
2. All antennas satisfy the far-field criterion,
3. Precise boresight alignment between transmitting and receiving antennas,
4. Good impedance matching and polarisation match,
5. Negligible proximity effects and multipath interference.

Common sources of error in such measurements include ground reflections, scattering from nearby objects, residual multipath, and antenna misalignment. These effects can be minimised by:

- Maintaining sufficient separation distance ($R \gg 2D^2/\lambda$),
- Using RF absorbers to suppress unwanted reflections,
- Performing measurements at multiple distances and averaging the results,
- Employing high-quality cables and connectors with careful VNA calibration.

Reasons for inaccuracies are discussed in detail in [19] and [5].

6.6 Comparison with Simulation and Discussion of Discrepancies

Parameter	Simulated	Measured
Resonant Frequency (GHz)	2.27	2.153
-10 dB Bandwidth (MHz)	140	132
Realised Gain (dBi)	5.53	6.353
HPBW E-plane (°)	27.4	15
Side-Lobe Level (dB)	-5.6	-6.95

Table 5: Comparison of simulated and measured performance.

Overall, good agreement is observed between the simulated and measured results, validating the effectiveness of the proposed tapered corporate feed microstrip patch antenna design.

The primary objective of this work was to develop a microstrip patch antenna operating at 2.2 GHz with sufficient bandwidth to cover the spacecraft downlink frequency band (2200–2290 MHz) for distances up to 2 million km from Earth. The measured performance largely meets this requirement.

The most noticeable discrepancy is the downward shift in resonant frequency by approximately 117 MHz (2.27 GHz simulated vs. 2.153 GHz measured). This shift is primarily attributed to the difference between the nominal dielectric constant used in simulation ($\varepsilon_r = 4.3$) and the actual measured value of the fabricated FR4 substrate ($\varepsilon_r = 4.8$). In future iterations, it is strongly recommended to characterise the substrate permittivity and loss tangent prior to simulation, or to use a more stable and homogeneous substrate such as Rogers RO4003C.

The measured -10 dB bandwidth of 132 MHz comfortably covers the required 90 MHz downlink band with a 42 MHz margin. The slight reduction compared to the simulated 140 MHz is typical and can be explained by fabrication tolerances (etching accuracy, substrate permittivity variation, and copper roughness), dielectric losses, and minor effects from the feed network.

The measured realised gain exceeds the simulated value by 0.8 dB. This improvement can be attributed to several factors, including constructive interference from residual ground reflections and multipath in the measurement chamber, as well as fabrication and material variations that resulted in slightly higher radiation efficiency. A particularly significant contributor is the effect of the SMA connector and RG316 coaxial cables, which act as parasitic ground extensions. The outer braid of the coax cable is electrically connected to the ground plane, effectively increasing its size beyond that modelled in simulation.

A larger effective ground plane reduces edge diffraction and truncation currents, causing less energy to spill into wide angles. Consequently, the main beam becomes sharper, leading to higher directivity and peak gain [20].

This is clearly evidenced by the measured E-plane half-power beamwidth (HPBW), which is significantly narrower than simulated (15° versus 27.4°). The reduced beamwidth directly corresponds to the observed increase in realised gain.

The measured side-lobe level is marginally better than simulated (-6.95 dB versus -5.6 dB), indicating improved rejection of interfering signals in unwanted directions.

These discrepancies are typical in low-cost FR4 prototypes and arise from a combination of real-world effects (connector/cable parasitics, measurement environment, and material variations) that are difficult to model perfectly. Despite the differences, the measured results remain well within acceptable limits and confirm the validity of the antenna and feed structure design.

In conclusion, the close correlation between simulation and measurement, combined with compliance to the target bandwidth and gain requirements, demonstrates the success of the proposed antenna.

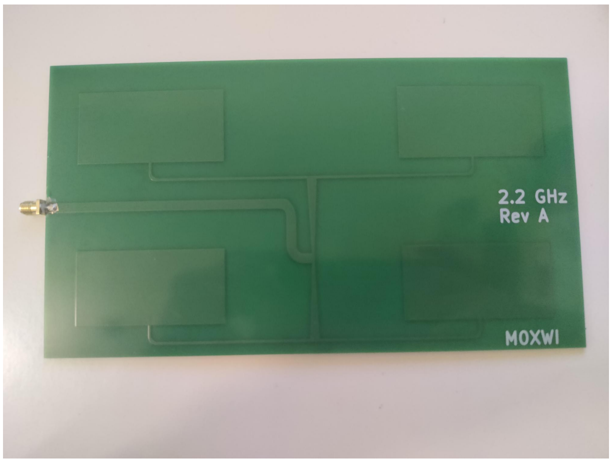


Figure 25: Fabricated Microstrip Patch Antenna Array

7 Conclusions

7.1 Summary of How the Design Meets the Requirements

The fabricated 2×2 tapered corporate-fed microstrip patch array successfully meets the core requirements: it resonates near 2.2 GHz, provides more than 130 MHz of -10 dB impedance bandwidth (comfortably covering the 2200 MHz to 2290 MHz LEO downlink band), and achieves a measured realised gain of approximately 6 dBi. The use of linear tapered impedance transformers demonstrably widens the matching bandwidth compared with conventional quarter-wave designs. The main trade-off is the moderate gain and efficiency imposed by the lossy FR4 substrate; higher performance would require a lower-loss laminate. For production or higher-gain systems, a Rogers RO4003C (or equivalent) substrate is recommended.

7.2 Recommendations for Future Work

- Fabricate a second prototype on Rogers Substrate to ensure homogeneous dielectric constant
- Extend the design to a 4×4 array for higher directivity.
- Move to antenna in package design to reduce interconnect losses
- Perform outdoor far-field measurements or use a proper anechoic chamber to quantify multipath effects.

7.3 Reflection

In deciding the project, we wanted to go with an amateur radio project. This was a good project to build fundamentals in RF engineering, however what could make the project even more useful would be a comprehensive state-of-the-art overview based on economic principles and understand how many people it would impact. An idea for a useful project would be to understand the limiting factors, and from that determine the requirements of the project that would be needed to solve the limiting factors. Performing this decision-making based on economic principles, talking to those who are working on the state of the art, will contribute a better project overall.

Secondly, regarding projects it should be a priority to get to the minimum viable product as soon as possible. It is of high importance to avoid vacillating endlessly over decisions and deliver a minimum viable product, and then perform AB testing with the minimum viable product. I found that I spent a lot of my time on learning about different types of antennas such as the vivaldi, and fractal, but with no real idea of how I was going to turn them into something useful. To solve difficult problems and work on challenging projects, you must set aggressive deadlines, and deliver a minimum viable product as soon as possible.

Furthermore, when working on anything that matters, it is useful to begin from first principles, working your way up from what you know to be axioms and then building on top of those axioms, rather than learning from analogy, working from analogy usually only leads to incremental improvements.

Throughout, writing and documentation must be prioritised, writing is one of the best ways to determine if you understand the concept. If you cannot write something in your own words, then you don't understand it. Writing is nature's way of showing you how well you understand a concept. Everything we do is for the sake of writing.

Humans have cognitive biases, namely that we prefer to disregard information that doesn't agree with our beliefs, and prefer information that correlates with our beliefs, this leads to delusional thinking that is not based on truth. It is critical to close the loop on reality, seek information that may contradict what we think we know, and be continuously developing. You have to be hyper-focused on the truth, and close the loop between ego and ability.

It is also important to not overpromise on projects, and ensure that you deliver what you set out to, and if you can achieve that is great. Trust is an integral part of being part of organisation, and overpromising on a project is one of the fastest ways to lose that trust.

Another point of reflection was that the future is a concept, only the present is now, therefore it is very important to not overly forecast, it is important to plan ahead, while also staying in the present moment, the only moment that is real is the present moment and dealing with the present moment and what you are able to do in the present gives you far more control.

Regrets are a tax on your mind. Move forward and do the best you can right now. Perform error correction on your mistakes and learn from them.

It is also not a question not a question of only how hard you work, but also if you are working in the right direction, you must often have to perform course corrections, if some new information has surfaced.

Humans are cybernetic, which means that a team is its vector sum, therefore, you are only as strong as your weakest link.

In regards to technical content, this project has been very useful for building an understanding of electromagnetics, antennas, fabrication, and and instrumentation.

In regards to self-development, the project has given me the opportunity to tackle a real engineering project, understand its potential pitfalls and learn from those errors.

References

[1] S. Khan, T. Mazhar, T. Shahzad, A. Bibi, W. Ahmad, M. A. Khan, M. M. Saeed, and H. Hamam, “Antenna systems for IoT applications: a review,” *Discover Sustainability*, vol. 5, no. 1, p. 412, 2024.

[2] R. Garg, P. Bhartia, I. Bahl, and A. Ittipiboon, *Microstrip Antenna Design Handbook*. Norwood, MA: Artech House, 2001.

[3] Deep Space Network and D. K. Shin, “Frequency and channel assignments,” Jet Propulsion Laboratory, California Institute of Technology, DSN Telecommunications Link Design Handbook 810-005, 201, Rev. F, May 2022, https://deepspace.jpl.nasa.gov/dsndocs/810-005/201/201F.pdf.

[4] *IEEE Standard for Definitions of Terms for Antennas*, IEEE Std. IEEE Std 145-2025 (Revision of IEEE Std 145-2013), 2025.

[5] C. A. Balanis, *Antenna Theory: Analysis and Design*, 4th ed. John Wiley & Sons, 2016.

[6] D. M. Pozar, *Microwave Engineering*, 4th ed. Hoboken, NJ: Wiley, 2012.

[7] R. J. Mailloux, *Phased Array Antenna Handbook*, 3rd ed. Artech House, 2018.

[8] E. Hammerstad and O. Jensen, “Accurate models for microstrip computer-aided design,” in *1980 IEEE MTT-S International Microwave Symposium Digest*, 1980, pp. 407–409.

[9] M. Steer, “7.5: Tapered matching transformers,” May 2022. [Online]. Available: https://eng.libretexts.org/Bookshelves/Electrical_Engineering/Electronics/Microwave_and_RF_Design_III_-_Networks_(Steer)/07%3A_Chapter_7/7.5%3A_Tapered_Matching_Transformers

[10] S. J. Mumby and J. Yuan, “Dielectric properties of FR-4 laminates as a function of thickness and the electrical frequency of the measurement,” *Journal of Electronic Materials*, vol. 18, no. 2, pp. 287–292, 1989.

[11] A. Djordjevic, R. Biljie, V. Likar-Smiljanic, and T. Sarkar, “Wideband frequency-domain characterization of FR-4 and time-domain causality,” *IEEE Transactions on Electromagnetic Compatibility*, vol. 43, no. 4, pp. 662–667, 2001.

[12] Adrian, “Boosting signal integrity: A deep dive into surface finish selection for RF PCBs,” Jun. 2025. [Online]. Available: https://www.allpcb.com/blog/pcb-knowledge/boosting-signal-integrity-a-deep-dive-into-surface-finish-selection-for-rf-pcbs.html

[13] C. Riener, T. Bauernfeind, S. Kvasnicka, K. Roppert, H. Hackl, and M. Kaltenbacher, “Numerical investigation of signal launch imperfections for edge mount rf connectors,” *Electronics*, vol. 11, no. 13, 2022.

[14] T. Hirano, J. Hirokawa, and M. Ando, “Influence of the SMA connector and its modeling on electromagnetic simulation,” *Microwave and Optical Technology Letters*, vol. 57, no. 9, pp. 2168–2171, 2015.

[15] “Keysight fieldfox application note,” https://www.keysight.com/gb/en/assets/7018-03479/application-notes/5991-0421.pdf, accessed: 2026.

[16] Altium, "Differences between prepreg material vs. pcb core: What designers need to know," Jan. 2020. [Online]. Available: https://resources.altium.com/p/pcb-core-vs-prepreg-material-what-designers-need-to-know

[17] *IEEE Recommended Practice for Antenna Measurements*, IEEE Std. IEEE Std 149-2021 (Revision of IEEE Std 149-1977), Feb. 2022.

[18] A. Repjar, A. Newell, and D. Tamura, "Extrapolation range measurements for determining antenna gain and polarization," National Institute of Standards and Technology, Technical Note 1311, 1987.

[19] R. W. Beatty, "Discussion of errors in gain measurements of standard electromagnetic horns," National Institute of Standards and Technology, Technical Note 351, 1967.

[20] J. Huang, "The finite ground plane effect on the microstrip antenna radiation patterns," *IEEE Transactions on Antennas and Propagation*, vol. 31, no. 4, pp. 649–653, 1983.